ESP32 Firmware Development Instructions

Objective

Develop the ESP32 firmware with four operational modes. The system will simulate gestures **Arrays** using the **Serial Monitor** and perform the appropriate actions for each mode. This document provides a comprehensive breakdown of the development process, including the four Modes.

System Overview

1. **ESP32**:
   Acts as the central controller, communicating with the Palm PCB via UART and emulating HID devices (mouse, gamepad, VR controller) via Bluetooth.
2. **Palm PCB**:
3. **Serial Monitor**:
   Used to simulate gestures by entering numbers (1 to 6), representing gestures (Arrays 1 to 6).
4. **Modes**:
   - **Mode 1**: Palm PCB Control
   - **Mode 2**: Mouse Control
   - **Mode 3**: Gamepad Control
   - **Mode 4**: VR Control
5. **LED Indicators**:
   Used to provide feedback on the active mode.
6. **Mode Button**:
   Cycles through the four modes.

Development Instructions

1. General Setup

- **UART Configuration**:
  Set up UART on the ESP32 to communicate with the Palm PCB at a baud rate of **38400**.
- **Bluetooth HID Configuration**:
  Configure the ESP32 to act as a Bluetooth HID device for mouse, gamepad, and VR modes.
- **Serial Monitor Input**:
  Set up the ESP32 to read numbers (1 to 6) from the Serial Monitor to simulate gestures.
- **Mode Management**:
  Implement a state machine to handle mode switching using a Mode Button.
  LED Blink Patterns for Modes:
  - **1 Blink**: Mode 1 (Palm PCB Control)
  - **2 Blinks**: Mode 2 (Mouse Control)
  - **3 Blinks**: Mode 3 (Gamepad Control)
  - **4 Blinks**: Mode 4 (VR Control)

Check Below Modes Instructions

Mode 1: Palm PCB Control Mode

Objective

Send Serial commands (G1–G6) to the Palm PCB via UART based on simulated gestures.

Steps

1. **Setup UART Communication**
    - Baud rate: **38400**.
    - Connect:
        - **ESP32 TX → Palm PCB RX**
        - **ESP32 RX → Palm PCB TX**
        - **GND → GND**
2. **Simulate Gestures via Serial Monitor**
    - Input numbers (1 to 6) in the Serial Monitor to simulate gestures.
3. **Command Mapping**
    - Map gestures to grip commands:

Input (Array)	Command	Description
1	G1	FIST
2	G2	PALM
3	G3	THUMBS
4	G4	POINT
5	G5	PINCH
6	G6	TRIPOD

4. **Send Commands to Palm PCB**
    - Transmit the corresponding command over UART.
5. **Error Handling**
    - Handle invalid inputs (e.g., 7, a) by displaying an error message in the Serial Monitor.
    - Retry UART transmission up to 3 times on failure.
6. **LED Feedback**
    - Blink the LED **once** to indicate Palm PCB Control mode.

Mode 2: Mouse Control Mode

Objective

Perform mouse actions (e.g., Left Click, Scroll) based on Simulated Array and output results via Bluetooth.

Steps

1. **Configure ESP32 as a Bluetooth HID Device**
    - Set up the ESP32 to function as a Bluetooth mouse.
2. **Simulate Gestures via Serial Monitor**
    - Input numbers (1 to 6) in the Serial Monitor to simulate gestures.
3. **Mouse Action Mapping**
    - Map gestures to mouse actions:

Input (Array)	Mouse Action	Description
1	Left Click	Perform left-click
2	Right Click	Perform right-click
3	Double Click	Perform double-click
4	Scroll Up	Scroll up on screen
5	Scroll Down	Scroll down on screen
6	Middle Click	Perform middle-click

4. **Perform Mouse Actions**
    - Send the corresponding mouse action via Bluetooth.
5. **Error Handling**
    - Display an error message for invalid inputs.
    - Handle Bluetooth disconnections and attempt automatic reconnection.
6. **LED Feedback**
    - Blink the LED **twice** to indicate Mouse Control mode.

Mode 3: Gamepad Control Mode

Objective

Simulate gamepad button presses based on simulated gestures and output results via Bluetooth.

Steps

1. **Configure ESP32 as a Bluetooth HID Gamepad**
    - Set up the ESP32 to function as a Bluetooth gamepad.
2. **Simulate Gestures via Serial Monitor**
    - Input numbers (1 to 6) in the Serial Monitor to simulate gestures.
3. **Gamepad Button Mapping**
    - Map gestures to gamepad buttons:

Input (Array)	Gamepad Button	Description
1	Cross	Select/Interact
2	Circle	Back/Cancel
3	Triangle	Special action
4	Square	Secondary action
5	R1 Button	Shoulder button
6	R2 Button	Trigger button

4. **Perform Gamepad Actions**
    - Send the corresponding gamepad action via Bluetooth.
5. **Error Handling**
    - Display an error message for invalid inputs.
    - Handle Bluetooth disconnections and attempt automatic reconnection.
6. **LED Feedback**
    - Blink the LED **three times** to indicate Gamepad Control mode.

Mode 4: VR Control Mode

Objective

Simulate VR controller actions based on simulated Arrays (serial) and output results via Bluetooth.

Steps

1. **Configure ESP32 as a Bluetooth HID VR Controller**
    - Set up the ESP32 to function as a VR controller.
2. **Simulate Gestures via Serial Monitor**
    - Input numbers (1 to 6) in the Serial Monitor to simulate gestures.
3. **VR Controller Button Mapping**
    - Map gestures to VR actions:

Input (Array)	VR Action	Description
1	A Button	Select/Interact
2	B Button	Back/Cancel
3	Right Trigger	Grab/Shoot
4	Right Grip Button	Grip/Hold
5	Menu Button	Access menu
6	Oculus Button	VR Dashboard

4. **Perform VR Actions**
    - Send the corresponding VR action via Bluetooth.
5. **Error Handling**
    - Display an error message for invalid inputs.
    - Handle Bluetooth disconnections and attempt automatic reconnection.
6. **LED Feedback**
    - Blink the LED **four times** to indicate VR Control mode.

Check Below Ideal Operation Procedure

Ideal Operation Procedure

1. **Setup**:
    - Connect the ESP32 to the PC via USB and open the Serial Monitor (baud rate: 115200).
    - Set up necessary hardware connections (UART for Palm PCB, Bluetooth for other modes).
2. **Mode Selection**:
    - Use the Mode Button to cycle through the modes.
    - Verify active mode using LED blink patterns.
3. **Simulate Gestures**:
    - Enter numbers 1 to 6 in the Serial Monitor.
    - Verify the corresponding output or action based on the active mode.
4. **Error Testing**:
    - Test with invalid inputs and verify error handling.
    - Test for disconnections and reconnection functionality in Bluetooth modes.